

Dhruv Singh

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EDUCATION

Tribhuvan College, Nalanda University Centre

Bachelor of Science (Hons.) in Data Science

Neemrana, Rajasthan

2023 – Present

- GPA: 7.75 (10 scale)
- Coursework: Machine Learning, Data Visualization, SQL, Python Programming

Symbiosis Public School

Senior Secondary (Class XII) PCM with Informatics Practices

Dhanbad, Jharkhand

2021 – 2023

- Percentage: 74%

EXPERIENCE & RESEARCH

OSCode

Web Development Intern

Noida, Uttar Pradesh

2026

- **Architected & Developed Full-Stack Features:** Built responsive web interfaces using Next.js, React, and Tailwind CSS, and developed cross-platform mobile app screens using React Native (Expo).
- **Scalable Backend Engineering:** Engineered robust REST APIs and microservices using NestJS and Node.js, leveraging Prisma ORM with PostgreSQL for efficient data management.
- **Performance & Background Processing:** Implemented asynchronous background job processing and message queues using BullMQ and Redis to handle event registrations, notifications, and scalable mentor payouts.
- **Advanced Monorepo Management:** Maintained a complex Turborepo monorepo environment with shared TypeScript packages (UI, schemas, analytics), improving code reusability and CI/CD build times across 9+ applications.
- **Search & AI Integrations:** Integrated Typesense for lightning-fast mentor discoverability/search, and utilized the OpenAI SDK for advanced backend automations.
- **State Management & Validation:** Ensured type safety and robust form handling across the stack using Zod, React Hook Form, Zustand, and TanStack React Query.

Subharti University

Paper Presentation

Remote/Hybrid

2024

- Authored and presented a research paper titled “A Modern Way to Generate Electricity: Floating Solar Farms”.
- Analyzed the efficiency and environmental impact of floating solar photovoltaic systems compared to land-based alternatives.

Codtech IT Solutions

Python Hackathon Participant

Remote

2024

- Competed in a rigorous Python hackathon focusing on algorithmic efficiency and problem-solving.
- Developed scripts to solve data processing challenges within strict time limits.

PROJECTS

ShramSetu — Migrant Worker Presence Platform — *Next.js 16, React 19, TypeScript, Prisma, PostgreSQL, d3-geo, Gemini API*

github.com/dhruv9097/shram-setu

- Built solo for Smart India Hackathon PS1 (Ministry of Labour & Employment): an offline-first PWA that lets India’s 31.89 crore unorganised workers mark their own presence as they migrate, replacing eShram’s static home-address record with portable, verifiable proof of work.
- Engineered three degrading check-in channels — worksite QR scan, one-tap geofenced attestation, and a missed-call IVR path for feature phones — backed by an IndexedDB offline queue that drains on reconnect without rewinding a worker’s current location.

- Designed a privacy kernel enforcing k-anonymity ($k = 10$) with complementary suppression, 7-day coordinate coarsening to district granularity, and individual access gated behind a recorded legal basis and a worker-readable audit log; verified by 22 automated privacy assertions.
- Shipped an officer dashboard rendering a 726-district choropleth and 14 migration corridors from a bundled 179 KB TopoJSON set, requiring no tile server, plus an employer console issuing printable worksite QR codes and attested muster rolls.
- Constrained AI to schema-bound classification: Gemini triages multilingual grievances and translates relief broadcasts while the worker's verbatim words are preserved, with an icon-based fallback that keeps the flow working when no model is reachable.

Social Media Automation Engine — *Python, Gemini Vision API, Pollinations.ai, BeautifulSoup*

- Engineered a 7-phase multi-agent AI system to automate social media management spanning competitor intelligence, trend analysis, content creation, and AI image generation.
- Built specialized agents: Spy Agent (7-day competitor scraping), Auditor Agent (gap analysis), Image Analyst (Gemini Vision API), Trend Hijack Agent (world trend scoring), Architect Agent (thread/reply drafts), Engagement Agent (Golden Hour protocol), and Image Generator (Pollinations.ai).
- Architected a live/mock dual-mode pipeline with quota management, structured JSON reporting, and a Next.js social manager UI dashboard.

GitHub Trending Dashboard — *Python, BeautifulSoup, Pandas, Next.js, TypeScript*

- Built a full-stack dashboard that scrapes GitHub's daily trending page, computes a custom Hype Score (1.25x daily star growth), and surfaces insights in real time.
- Python backend uses BeautifulSoup and Pandas to parse, clean, and export repo data to CSV; Next.js/TypeScript frontend renders an interactive leaderboard.
- Designed a clean data pipeline from raw HTML scrape to a structured, auto-refreshing frontend view.

Physio-Detection System: Virtual Paint — *Python, OpenCV, MediaPipe*

- Developed a computer vision application allowing users to draw on-screen using hand gestures without physical contact.
- Leveraged MediaPipe for real-time hand-tracking and OpenCV for image processing and gesture rendering.
- Optimized gesture recognition algorithms to reduce latency and improve drawing precision.

TECHNICAL SKILLS

- **Languages:** TypeScript, Python (Intermediate), C++ (Basic), SQL (Basic)
- **Frontend & Mobile:** Next.js, React, React Native (Expo), Tailwind CSS, PWA / Service Workers, d3-geo
- **Backend & Databases:** Node.js, NestJS, PostgreSQL, Prisma ORM, Redis, SQLite, IndexedDB
- **Libraries & State Management:** Zustand, TanStack React Query, React Hook Form, Zod, OpenCV, MediaPipe, Pandas, NumPy, BeautifulSoup
- **Architecture & Tools:** Turborepo, BullMQ, Typesense, Offline-First Sync, k-Anonymity & Data Privacy, QGIS, Git, GitHub
- **AI & APIs:** Gemini API (Vision & Text), OpenAI SDK, Pollinations.ai, Multi-Agent Systems, Prompt Engineering
- **Domains:** Full-Stack Web/Mobile Development, Data Analysis, Data Visualization, Machine Learning, Computer Vision, AI Automation

LEADERSHIP & EXTRACURRICULAR

Tribhuvan College

Event Organizer

Neemrana, Rajasthan

2023

- Coordinated the "Principal's Meet," a summit discussing "Changing Climate & Environment" for local school principals.
- Organized an inter-school debate competition, managing logistics and participant coordination.